# Project Initialization and Planning Phase

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| Date | 03 July 2026 |
| Team ID | XXXXXX |
| Project Title | Smart-Lender-Loan-Eligibility-Prediction |
| Maximum Marks | 5 Marks |

**Project Proposal (Proposed Solution) report**

The Smart Lender project is a machine learning-based web application developed to predict the loan eligibility of applicants. The system analyzes applicant information such as income, education, employment status, loan amount, credit history, and property area to determine whether a loan should be approved or rejected. Using a trained machine learning model integrated with a Flask web application, the solution provides quick, accurate, and reliable predictions, reducing manual effort and improving the loan evaluation process.

| **Project Overview** | |
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| Objective | To develop a machine learning-based loan eligibility prediction system that accurately determines whether an applicant is eligible for a loan based on financial and personal details. |
| Scope | The project focuses on predicting loan eligibility using historical applicant data and machine learning algorithms. It includes data preprocessing, model training, prediction, and deployment through a Flask web application. The system is intended for educational purposes and can be extended for real-world banking applications. |
| **Problem Statement** | |
| Description | Banks receive numerous loan applications every day. Manual verification of applicant details is time-consuming and may lead to delays and inconsistent decisions. An automated system is required to assist in predicting loan eligibility quickly and accurately. |
| Impact | The Smart Lender system reduces loan processing time, improves prediction accuracy, minimizes manual effort, and supports faster decision-making for both applicants and financial institutions. |
| **Proposed Solution** | |
| Approach | The system collects applicant details through a web interface, preprocesses the data, and passes it to a trained machine learning model. The model predicts whether the loan should be approved or rejected, and the result is displayed instantly through the Flask application |

**Resource Requirements**

| **Resource Type** | **Description** | **Specification/Allocation** |
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| **Hardware** | | |
| Computing Resources | CPU/GPU specifications, number of cores | T4 GPU |
| Memory | RAM specifications | 8 GB |
| Storage | Disk space for data, models, and logs | 1 TB SSD |
| **Software** | | |
| Frameworks | Python frameworks | Flask |
| Libraries | Additional libraries | scikit-learn, pandas, numpy, matplotlib, seaborn |
| Development Environment | IDE | Jupyter Notebook, pycharm |
| **Data** | | |
| Data | Loan Eligibility Dataset | CSV format containing applicant details used for training and testing the machine learning model |